Clients & Ingress

Customer Web/Mobile App *[Container: React / React Native]* Provides UI for browsing flash catalog, locking items, and executing checkout.

Admin Web Portal *[Container: Angular / React]* Provides dashboard to schedule sales, set stock counts, and monitor live metrics.

HTTPS / WSS

SALESTORM Platform (Container Boundary)

Edge Infrastructure

CDN & DDoS Edge *[Container / Edge Worker: Cloudflare]* Caches static assets; routes high-frequency catalog read traffic.

HTTPS

HTTPS

API Gateway & Reverse Proxy *[Container: Kong / NGINX Envoy]* Rate limits, authenticates JWTs, routes API calls, checks idempotency keys.

gRPC / HTTPS

Application Containers (Kubernetes Engine)

gRPC / HTTPS

gRPC / HTTPS

gRPC / HTTPS

gRPC / HTTPS

Order Processing Service *[Container: Java Spring Boot]* Orchestrates order state machine and checkout workflows.

Verify Reservation

Auth & User Service *[Container: Node.js / Go]* Manages user accounts, sessions, and authentication tokens.

Catalog Service *[Container: Go / Java Spring]* Serves flash sale event details and active product listings.

Inventory Reservation Service *[Container: Go / Rust]* Executes sub-millisecond atomic stock reservation with 10-min TTL logic.

Payment Service *[Container: Node.js / Go]* Handles payment intents, idempotency checks, and gateway webhooks.

Read / Write Session

Atomic Lua / TTL Keys

Sync Check / Update

Publish Reserved Event

Publish Payment Event

Datastores & Event Bus

Redis Cluster *[Container: Redis 7.x In-Memory Store]* Holds atomic stock counters, 10-min TTL keys, and idempotency locks.

SQL Queries

Event Streaming Bus *[Container: Apache Kafka / Redpanda]* Buffers inventory-reserved, order-status, and rollback event streams.

Process Charge / Webhooks

Batch Outbox Writes

Trigger Alerts

Fulfill Orders

Transactional Database *[Container: PostgreSQL Distributed]* Stores persistent users, product metadata, orders, and payment records.

External Third-Party APIs

Payment Gateway *[External API: Stripe / Razorpay]*

Notification Gateway *[External API: Twilio / SendGrid]*

Logistics Provider *[External API: Shiprocket / FedEx]*